

Nicholas Johnson

njohns59@gmu.edu | 703-380-2204 | [Portfolio](#) | [LinkedIn](#) | [Github](#)

EDUCATION

George Mason University

Expected Summer 2027

B.S. Neuroscience, Concentration: Computational Neuroscience | Fairfax, VA

- Selected for GEO Neurotechnology in Germany Program, a 6-credit program visiting 8 research labs across Germany.

RESEARCH

NeuroMorpho.org Research Assistant → Research Intern (Aug. 2026)

June 2026–Present

Krasnow Institute for Advanced Study, George Mason University, Fairfax, VA | [NeuroMorpho.org](#)

- Contributed 240 edited neuron reconstructions to NeuroMorpho.Org's public database.

Literature Mining

- Conduct literature mining to identify neuronal and glial reconstructions for NeuroMorpho.Org database curation.
- Evaluate reconstruction evidence using tracing-system keywords including Neurolucida, Imaris/Filament Tracer, Amira, Vaa3D, Simple Neurite Tracer, and CATMAID.
- Curate morphology metadata by verifying article details, reconstruction counts, figure evidence, supplementary files, data availability, and DOI/PMID references.
- Review repositories including GitHub, Dryad, Figshare, Zenodo, and Dataverse for downloadable reconstruction data.

Neuron Morphology Editing

- Convert neuron reconstruction files submitted by labs worldwide into a standardized SWC format using autoprocessing; use Fiji and Neuromantic for files that require manual conversion.
- Fix structural errors in CVAPP: identify and mark somas, create single-point somas, reposition misplaced somas in 3D space, reconnect broken segments, fix long connections, repair incorrect topology, and label apical and basal dendrites and axons in pyramidal neurons.
- Troubleshoot file-saving issues using NeuTube and node-resampling.
- Run every file through an automated quality-control check that flags leftover errors, then fix and recheck until the reconstruction is clean and ready to upload.

Undergraduate Research Assistant

January 2026–May 2026

Anatomy and Physiology Education Research, George Mason University | Fairfax, VA

- Collaborated on manuscript preparation for a comparative anatomy and physiology education study.
- Contributed to data interpretation and formation of graphs in google colab for peer-reviewed publication.

PROJECTS

Hodgkin–Huxley Simulator

May 2026–Present

Independent Project | [hhsimulator.com](#)

- Built a cross-platform desktop application (Windows, macOS, Linux) implementing the Hodgkin–Huxley equations to model action potential generation and membrane dynamics in real time.
- Simulates single-compartment membrane dynamics with live visualization of membrane voltage, injected current, net ionic current, conductances (Na^+ , K^+ , leak), and gating variables (m , h , n).
- Validated outputs against the NEURON Simulator; authored a research-style manuscript covering methods, error analysis, biological interpretation, and future extensions.

WORK EXPERIENCE

AI Trainer / LLM Evaluator

2026–Present

Handshake AI Fellowship | Remote

- Evaluated large language model responses for accuracy, reasoning quality, instruction-following, safety, and overall usefulness.
- Provided structured feedback on prompts and outputs to help improve AI assistant behavior across varied tasks and domains.

MRI Office Assistant

July 2025–Present

Institute for Biohealth Innovation, George Mason University | Fairfax, VA

- Support research administration office operations and completed level one MRI training from a certified MRI Technologist.

Office Assistant, IET Department

February 2025–May 2025

Northern Virginia Community College | Annandale Campus, VA

- Built computational computer-network labs, supported hardware setup, laptop debugging, and switch configuration for NOVA students using AI supported tools.

TECHNICAL SKILLS

Programming & Tools
Modeling & Simulation
Neuroinformatics
AI
Cellular Neuroscience

Python, R, MATLAB, TypeScript, HTML, Git/GitHub, Tkinter (cross-platform desktop app development), Matplotlib, Google Colab
NEURON Simulator, Hodgkin–Huxley model, Cable equation
NeuroMorpho.Org, ModelDB, CVAPP, Neuromantic, NeuTube
Prompt development, LLM output review, Factual accuracy assessment, Reasoning quality assessment
Electrical properties of nerve cells, Synaptic transmission, Neurotransmitters and Receptors, Molecular signaling, Synaptic plasticity